



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI VENKATESWARA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

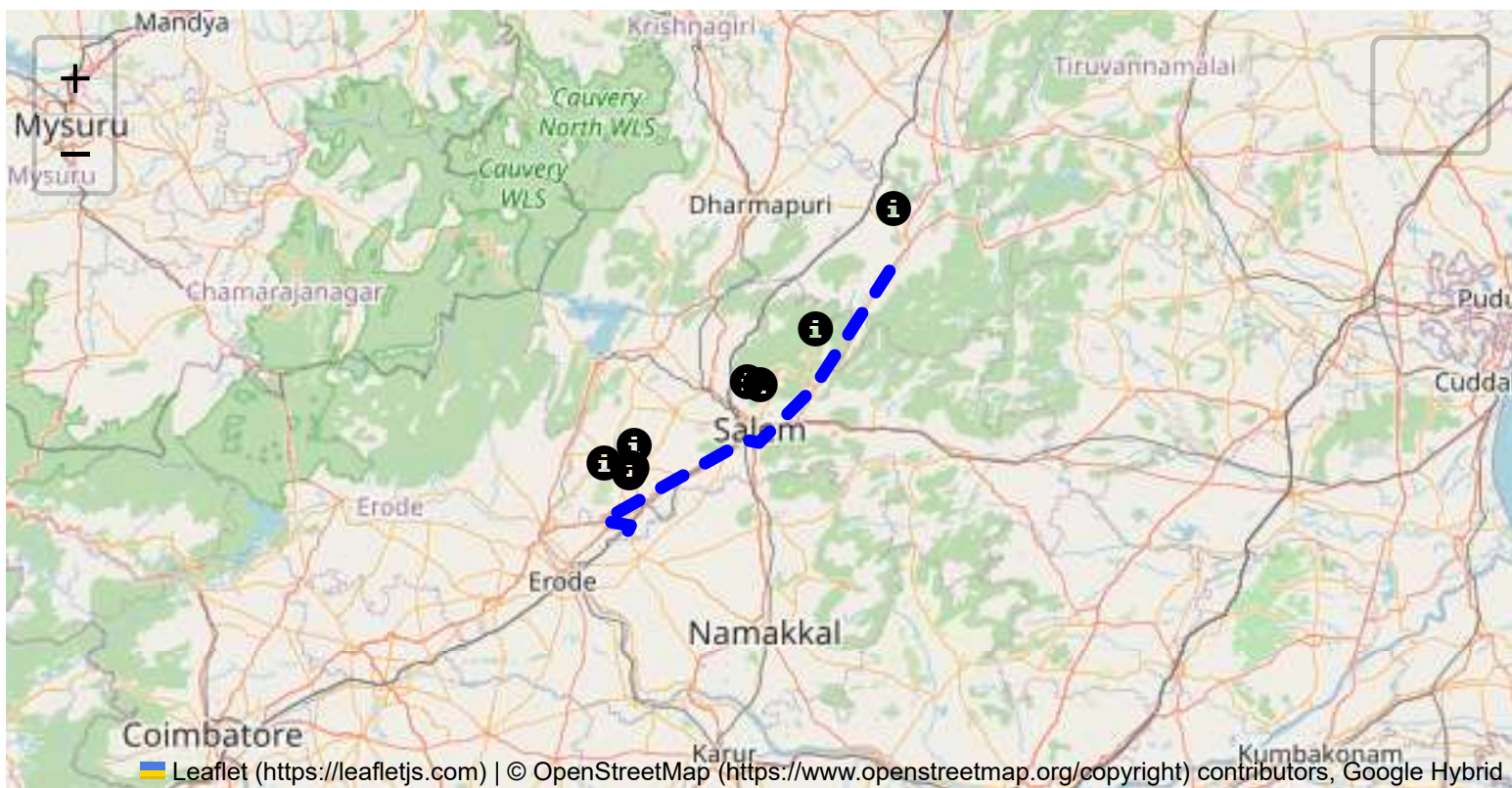
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 109.46 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.2 hours
Start: (11.4381, 77.8734)
End: (11.999973, 78.448311)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route covers approximately 109.46 kilometers starting from Sangagiri, Tamil Nadu, traveling through various localities including Padaiveedu and Salem, and ending at NH 179A near Gobinathampatti. It connects several highways including Salem - Kochi Highway and Kondalampatti Bypass.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** The region generally experiences hot and dry weather with a typical tropical climate. Monsoons can bring heavy rain from June to September.
- **Potential Hazards:** During monsoon seasons, heavy rains could cause flooding and poor visibility along the route, particularly in low-lying areas.

3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic congestion is highest during morning hours (8 AM - 10 AM) and evening hours (5 PM - 7 PM), especially in urban areas like Salem.
- **Congestion-Prone Areas:** The Kondalampatti Bypass and major intersections near Salem are known for heavy congestion.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Roads vary from well-maintained highways to narrower roads with potential potholes, especially in rural stretches.
- **Infrastructure:** The route has adequate signage and lighting in urban areas, but rural sections may lack sufficient lighting and clear signage.

5. Suggestions for Alternative Routes for Emergencies

- Use the Salem - Bangalore Highway as an alternative in emergencies for quicker access to major cities and facilities, though this may extend the travel time.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Time Restrictions:** Movement of heavy vehicles with hazardous materials might be restricted during night hours in urban zones.
- **Permits Required:** Drivers must ensure they have the necessary permits for hazardous materials, complying with Tamil Nadu's road safety regulations.

7. Overview of Historical Incidents

- Historical records indicate occasional incidents involving heavy vehicles due to poor visibility and uncontrolled intersections, particularly during rainy seasons.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Urban centers and ecologically sensitive areas near water bodies require careful navigation to avoid contamination.
- **Waste Management Zones:** Proper planning to avoid disrupting local waste management sites.

9. Analysis of Communication Coverage

- **Coverage:** Generally strong mobile network coverage in urban and semi-urban areas; however, rural stretches may have brief dead zones.
- **Note for Drivers:** Ensure emergency communication devices are charged and carry physical maps when traveling through rural areas.

10. Estimated Emergency Response Times

- **Urban Segments:** Estimated emergency response could be within 30 minutes.
- **Rural Segments:** Response times might extend to an hour or more based on proximity to the nearest town.

11. Overall Summary of Risk Assessment

- **Risks Identified:** Traffic congestion, potential weather disruptions from monsoons, variable road quality, and limited communication in rural sections.
- **Recommendations:** Plan travel during off-peak hours, monitor weather forecasts, stay informed on local traffic regulations, and carry necessary communication tools.

The analysis suggests careful planning and preparedness for adverse weather and traffic conditions, particularly when transporting hazardous materials on this route. Regular updates on road conditions, weather, and traffic advisories are vital for maintaining safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44898, 77.87410	30 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	Medium	11.45543, 77.81740	30 KM/Hr
9	Turn	High	11.45631, 77.81668	15 KM/Hr
10	Turn	Medium	11.64172, 78.11817	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Turn	Medium	11.64216, 78.11875	30 KM/Hr
12	Turn	Medium	11.64195, 78.12129	30 KM/Hr
13	Turn	High	11.64162, 78.12160	15 KM/Hr
14	Turn	Medium	11.64087, 78.12157	30 KM/Hr
15	Turn	Medium	11.63948, 78.12081	30 KM/Hr
16	Turn	Medium	11.62821, 78.13348	30 KM/Hr
17	Turn	Medium	11.62836, 78.16483	30 KM/Hr
18	Turn	High	11.66380, 78.20648	15 KM/Hr
19	Turn	High	11.67120, 78.23136	15 KM/Hr
20	Turn	High	11.67190, 78.23144	15 KM/Hr
21	Turn	Medium	11.67429, 78.23896	30 KM/Hr
22	Turn	Medium	11.81208, 78.34071	30 KM/Hr
23	Turn	Medium	11.83518, 78.35305	30 KM/Hr
24	Turn	Medium	11.99561, 78.44554	30 KM/Hr
25	Turn	High	11.99959, 78.44845	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
14	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
16	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
17	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
18	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
19	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
20	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
21	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
22	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
23	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
26	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
27	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
13	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
24	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
25	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64172, 78.11817



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64216, 78.11875



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64195, 78.12129



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64162, 78.12160



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64087, 78.12157



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63948, 78.12081



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.62821, 78.13348



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62836, 78.16483



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66380, 78.20648



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67120, 78.23136



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67190, 78.23144



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67429, 78.23896



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.81208, 78.34071



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83518, 78.35305



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99561, 78.44554



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.99959, 78.44845
